TOYLAND KB-09

User Manual

A recreation of Kevin’s Roland SH-09

Version 0.13.2 · Windows and unverified Mac test builds · 8 October 2026

A PERSONAL SYNTHESIZER PROJECT

A recreation of Kevin's Roland SH-09.

AGE lets you age both the instrument's appearance and its circuit-model behavior, from a fresh panel to yellowed keys, dust and worn metal. Shape a synth sound by hand, or send external audio through the filter and let the envelope follower move it with your playing.

UNOFFICIAL PERSONAL PROJECT — NOT AFFILIATED WITH ROLAND CORPORATION — NOT FOR SALE

Windows: extract the ZIP, then double-click INSTALL KB-09.cmd. Mac test build: extract its separate ZIP and open INSTALL KB-09.command. Full instructions follow.

Please send bugs and manual corrections to Kevin, the person who shared this instrument with you.

A personal synth, ready to explore

KB-09 brings Kevin's Roland SH-09 into a playable software instrument, with a hands-on panel modeled after his own aged synth. One oscillator, a sub oscillator, a resonant low-pass filter and a flexible envelope give you a direct path from a deep bass to a singing lead, a hollow pulse or a burst of noise percussion. Reach for a fader, switch a waveform and hear the sound take shape. Sixteen factory sounds offer starting points, and you can save every adjustment as your own preset.

AGE gives that instrument a condition of its own: the panel wears, the keys yellow and the circuit model's pitch calibration, filter response and envelope timing shift subtly. You can also bring in drums, guitar, vocals or another synth through External Input , shape them with the filter, and use the envelope follower to open it with the incoming signal's level. Play KB-09 in your DAW or open the standalone app for a session with just a MIDI keyboard.

Install & open

Windows VST3

1. Right-click the downloaded ZIP, select Extract All… , and open the extracted folder.

2. Close your music software. Double-click INSTALL KB-09.cmd , allow the Windows permission request, then press Enter when prompted. The installer copies all required files automatically.

3. Open your music software and find Toyland KB-09 under instruments. Rescan VST3 plug-ins in its settings if needed, then reopen the software.

Use a 64-bit Windows DAW with VST3 support. The installer places KB-09 in the standard Windows VST3 location and keeps any previous copy outside the plug-in scan folder.

Standalone

After installing, open the Windows Start menu, search for Play KB-09 , and open it. Its panel and factory sounds are embedded; no separate graphics installation is needed. Select the correct audio device and MIDI keyboard in the standalone settings. If another app is using an exclusive audio driver, close that app or choose a shared device.

Mac AU / VST3 — unverified test build

Download the separate Mac ZIP and extract it. Close your music software, then double-click INSTALL KB-09.command and press Return when prompted. Reopen your music software and rescan its plug-ins. Logic Pro and GarageBand use the AU; other compatible hosts can use the VST3. This installs for all users of your Mac, on macOS 12 or later, Intel or Apple Silicon. The installer asks for your Mac login password; typed password characters remain hidden.

The Mac test build is ad-hoc signed, not Developer ID signed or notarized. If macOS blocks it, contact Kevin for help; do not disable system security. Apple AU validation passed on the build runner after system-wide installation. This does not establish compatibility with your particular host. Please report your Mac, macOS and host versions when testing.

Which file do I need?

Format	Use it for	Availability
VST3	A compatible Windows DAW, including Live	Included; ready to install

Format	Use it for	Availability
Standalone EXE	Playing without a DAW	Included; ready to open
AU — Audio Unit	Mac hosts such as Logic Pro and GarageBand	Separate unverified Mac test download
AAX Native	Pro Tools	Included separately as an unsigned Windows development build; not ready for ordinary Pro Tools

Use the Windows VST3 for a compatible DAW, or the EXE to play on its own. AU is Apple's Mac plug-in format. The separate Mac test download contains AU and VST3 versions, marked unverified until tested in everyday Mac hosts. The separate AAX development build is unsigned and has not been tested in Pro Tools. It is for development testing only. Changing a file's extension does not change its format.

Start playing in a minute

1. Install the Windows VST3 , then load Toyland KB-09 on an instrument or MIDI track.

2. Choose 01 / Calibrated Init in the preset field along the top.

3. Turn down your listening level, enable the track's MIDI monitoring and play a MIDI keyboard. You can also click KB-09's keys.

4. Move CUTOFF FREQ to hear the sound darken or brighten. Add a little RESONANCE for a stronger filter peak.

5. Try 02 / Fundamental Bass or 04 / Singing Lead , then save a sound of your own .

KB-09 is monophonic : it plays one note at a time, with the lowest held note taking priority. Overlapping notes and the envelope trigger switch shape its legato behaviour. Click and hold an onscreen key, or hold a MIDI note, to see its blue highlight. A small amber bar along the key's lower edge identifies the note currently playing; other held keys keep a blue bar. The marks disappear when the keys are released.

Meet AGE

Choose the condition of your instrument.

CONDITION / AGE lives below the instrument. At 0 it selects the healthy calibration profile. Turn it up to explore a more played-in instrument, then turn it back whenever you like. AGE is saved with your sound and works in both Authentic and Modern modes.

The wear is modeled after Kevin's own aged SH-09. Dust settles in slider openings, around knob skirts and in grip ribs; keys warm to ivory and lose their shine; printing discolours; and worn panel paint reveals the metal beneath. The knobs' metal surfaces dull and develop fine pitting, while pale oxidation gathers on the screws. Explore the look at a larger size to see the small details.

What changes?

Appearance

Yellowing, dust, dulling and tarnish build gradually. Distinct stains appear separately and keep their shape once present, including a coffee ring in the envelope section. One key develops a fine hairline crack that can extend a little. Later wear can remove a fader cap or two and break the bender tip, revealing the parts beneath.

The controls continue to work through every condition. Lower AGE whenever you want to bring back the caps, intact bender and fresher panel.

Sound

AGE subtly changes pitch calibration, filter tuning/resonance and envelope timing within the circuit model. The same setting gives the same condition each time, so you can save it with a patch and return to it.

AGE is a creative condition control rather than a clock counting elapsed years. It does not add random crackles or make a worn key stop playing.

Try this

Save a favourite patch. Hold one note and move AGE slowly from fresh to worn. Compare the subtle sound changes, then explore the filter and envelope with AGE at a setting you like. Inspect the panel at L size, and load your saved patch whenever you want to return to the starting point.

Learn the panel



Drag sliders vertically; drag a knob vertically or horizontally. Hover or drag to read its value bubble. Most sound controls use a 0–10 display; TUNE displays cents. Double-click a continuous control to return to its default. Click or drag the ridged switches, or click their adjacent labels. Focused switches also accept arrow keys, Home and End.

Click the grey POWER button to switch the instrument on or off. It stands tall when OFF and sits lower when ON; the red lamp below the POWER label lights when the synth is on. Switching it off mutes the sound and clears held notes. Play a fresh note after switching it back on.

Section	What it does	A first experiment
MODULATOR	The low-frequency oscillator (LFO) moves pitch, filter or pulse width. Select sine, square or random/sample-and-hold; RATE sets speed and DELAY TIME delays its note-triggered modulation rise.	Add a little VCO MOD for vibrato. PWM uses the undelayed sine LFO.
VCO	The main oscillator selects noise, saw, square or variable pulse. RANGE changes octave; TUNE adjusts fine pitch. PULSE WIDTH MOD selects LFO, Manual or Envelope.	Select variable pulse, then raise PULSE WIDTH in Manual mode.
SUB OSC	A divider adds a square one octave down, square two octaves down, or narrow pulse two octaves down.	Raise SUB OSC in the mixer for a heavier bass.
AUDIO MIXER	Balances VCO, SUB OSC and the external audio input.	Lower VCO and raise SUB to hear the divider on its own.
VCF	The low-pass filter removes upper frequencies. CUTOFF sets brightness; RESONANCE emphasizes the cutoff region. MOD adds LFO motion; ENV adds positive envelope, inverted envelope or external-follower motion.	Lower cutoff, add ENV and use a short DECAY for a plucked tone.
VCA	HOLD passes sound continuously; ENV shapes level with ADSR; GATE follows held notes.	Use ENV for expressive note shapes. HOLD can keep sounding after keys are released.
ENVELOPE	A = rise time; D = fall to sustain; S = held level; R = fade after release. Gate + Trigger re-attacks when the effective note changes; Gate supports legato; LFO provides cyclic triggering while a note is selected.	Increase A and R for a soft entrance and tail.
PERFORMANCE BAY	VOLUME sets output level; PORTAMENTO glides between pitches. The bender returns to centre; its VCO and VCF sliders set separate sensitivities. POWER mutes the instrument.	Add portamento and overlap two notes. Play a fresh note after turning power back on.

Authentic & Modern

Authentic keeps the original-style routing and ignores the optional velocity, aftertouch, LFO tempo-sync and input-trim settings. Those controls dim. Modern enables them: velocity controls level, channel aftertouch opens the filter, TEMPO syncs the LFO, and INPUT TRIM adjusts external input by ±24 dB. Return to Modern to use their retained settings.

Modern provides deeper maximum LFO modulation: ±36 semitones for VCO pitch and ±8 octaves for the VCF, compared with Authentic's ±12 semitones and ±5 octaves. These are software design ranges. Modern sounds saved before 0.11 with nonzero MOD can sound deeper after upgrading. The bender sensitivities are separate.

The header's ENV and EXT indicators show envelope/follower activity. EXT AUDIO opens a routing reminder over the keyboard; click it again to close.

Save your sounds

1. Adjust the panel, including AGE and the Authentic/Modern choice.

2. Click SAVE , choose a location you can find again and give the sound a name. KB-09 writes a .kb09 file.

3. To recall it, click LOAD and select that file. Compatible earlier .sh09state files also load.

The loaded filename appears in the preset field. The sixteen factory presets remain available; selecting one recalls its original settings. Save your edits before choosing another sound. Cancelling a file dialog or rejecting an invalid preset leaves the current sound unchanged.

A preset saves the sound parameters, including AGE. It does not save held notes, the current envelope position, power or window placement. Your DAW also saves the patch with its project; external track routing belongs to that DAW project.

What is in the factory bank?

01 Calibrated Init · 02 Fundamental Bass · 03 Hollow Pulse · 04 Singing Lead · 05 Resonant Steps · 06 Soft PWM · 07 Low Drum · 08 Noise Snare · 09 Random Circuit · 10 Filter Whistle · 11 Sub Drone · 12 External Filter · 13 External Follower · 14 Played-in Bass · 15 Modern Expression · 16 Modern Clockwork.

External Filter and External Follower need routed audio. Random Circuit and Sub Drone use VCA Hold and can continue sounding without a held note.

A sharp, comfortable view

Window size and sharpness are separate. Use KB-09's own size buttons for comfort. If a host enlarges the plug-in as a bitmap, a bigger window can remain blurry; use the relevant DAW setting to allow native rendering.

Fits within about 65% of the available width and height.

Fits within about 90% of available width and height.

Requests the monitor's available work area.

The panel keeps its original proportions. Different screen shapes can leave dark margins. The maximise/restore button is at the top right. The standalone app supports dragging to the top edge to maximise; a DAW controls its own plug-in frame and may restrict movement or size requests.

Keep the display at its native resolution and retain the DAW's normal HiDPI support. Labels and drawn details stay sharp at the chosen size; photographic surfaces retain the detail of their source images.

Find your DAW

Choose your DAW for Windows display help. These instructions help KB-09 render at its own resolution while you use S/M/L to choose its size. Menu wording and available options vary by DAW version; vendor documentation is linked with each recipe.

Ableton Live

For a blurry KB-09 window on Windows, find the plug-in in Live's Browser, right-click it and disable Auto-Scale Plug-in Window , if offered. Kevin confirmed on 7 October 2026 that this fixed the display in his Live setup.

Then use KB-09's S, M or L button for size. If the change is not reflected, close/reopen the editor. If needed, save the patch and try a freshly inserted instance.

Option availability varies by Live version and plug-in format. Ableton's general description excludes VST3, although this toggle worked for Kevin's KB-09 setup. Ableton HiDPI guidance .

Ableton HiDPI guidance

Load KB-09 on a MIDI track, enable monitoring/arm as appropriate, and choose the keyboard as MIDI From. For audio processing, use Live external-input routing .

Cubase / Nuendo

Keep the DAW's global Enable HiDPI option on. Open the plug-in window's Functions menu, or its unused-space context menu. If present, leave Allow Window to be Resized unchecked so KB-09 can handle its own scale. Steinberg provides that option to help plug-ins without their own DPI handling.

Close/reopen the editor and try a fresh instance if needed. Use KB-09's own S/M/L buttons to choose a comfortable size.

See Cubase Pro 15 window options and Nuendo 14 window options . Menus can differ between versions and plug-in types.

Cubase Pro 15 window options

Nuendo 14 window options

Studio One / Fender Studio Pro

Retain the application's High-DPI mode. In the plug-in editor tab's settings menu, turn Enable System DPI Scaling off when offered. That option makes older small interfaces larger through bitmap scaling and can blur a plug-in that already handles its own scale.

Reopen the editor, and save your patch before trying a new instance if needed. For Studio One 7, search the setting name in Help > Studio One Reference Manual ; the current online manual is branded Fender Studio Pro 8.1, so its application menu name differs.

Official display options · How System DPI Scaling works .

Official display options

How System DPI Scaling works

FL Studio

Open the plug-in wrapper's additional settings and its Settings tab. Enable DPI aware so the plug-in handles its own scale. Keep KB-09's own size buttons as the normal sizing controls.

If the ordinary DPI-aware setting does not resolve the display, Image-Line also documents Processing > Make bridged plus DPI aware when bridged as a rescaling option. Save a wrapper favourite preset to retain GUI settings, then reopen the plug-in from it.

See the official Plugin Wrapper manual . Main GUI Scaling changes the whole DAW and requires a restart; start with the plug-in wrapper settings for a problem affecting KB-09 alone.

official Plugin Wrapper manual

REAPER

In Add FX , right-click KB-09's name and open Compatibility settings > HiDPI compatibility mode when floating windows . Leave that compatibility mode unchecked to let KB-09 render at its own scale.

The browser default applies when a new instance is added. Save the current patch, then insert a new instance and open its floating window. This setting applies to floating plug-in windows; use S/M/L inside KB-09 for size.

Official REAPER 7.81 guide , sections 6.10 and 6.19.

Official REAPER 7.81 guide

Bitwig Studio Version-dependent option

Windows versions have offered a per-plug-in automatic stretching choice. If your Browser context menu contains Stretch Plug-in Window to Match DPI , disable it for KB-09 and compare the result. Retain normal application/interface scaling.

The menu label can differ between versions. If the option is available, reopen the editor after changing it and try a fresh instance if needed.

Bitwig's Windows stretching option · The option in Bitwig 3.3.7 · Current plug-in guide .

Bitwig's Windows stretching option

The option in Bitwig 3.3.7

Current plug-in guide

Logic Pro / GarageBand / Pro Tools Mac testing / AAX unavailable

Logic Pro and GarageBand use Apple's Audio Unit (AU) format on Mac. Use the separate unverified Mac test download for AU installation. It has not been tested in Logic Pro or GarageBand. Pro Tools uses AAX ; KB-09 has no signed, host-tested AAX release. An unsigned development build cannot be treated as a plug-in ready for ordinary Pro Tools. Use the Windows VST3 in a compatible DAW, or the standalone app, for this release.

Apple Audio Unit compatibility · Avid AAX overview .

Apple Audio Unit compatibility

Avid AAX overview

Filter external audio

KB-09's VST3 instrument exposes an External Input auxiliary audio bus. A DAW may call this a sidechain. Here the incoming audio can become part of the sound, passing through the mixer, filter and amplifier.

1. Route an audio source to the instrument's External Input.

2. Choose 12 / External Filter . It lowers the oscillators, raises EXT and uses VCA Hold, so no MIDI note is needed.

3. Adjust CUTOFF FREQ and RESONANCE.

4. Try 13 / External Follower : louder input opens the filter. Adjust the VCF ENV amount and the source level.

The follower listens before the EXT mixer slider. Lowering EXT can make the source quieter while its filter modulation remains active. The follower does not trigger ADSR or open the VCA; use Hold for continuous processing. Stereo input is averaged to mono, so opposite-polarity channels can cancel. Output is mono duplicated to stereo.

In Ableton Live

1. Place KB-09 on a MIDI track; put the source audio on another track.

2. Expand the plug-in device's Sidechain panel. Enable Sidechain, choose the source under Audio From and select the intended tap, such as Post FX.

3. Set the sidechain mix to 100% for a known starting point, then choose External Filter in KB-09.

4. Monitor the processed path clearly. Prevent the source's direct output from masking the filtered result.

5. Save the Live Set to retain the track routing, then reopen it and verify the source and patch.

See the Live 11 sidechain routing guide . Menu names may vary in later versions. In another DAW, use its auxiliary-input or sidechain routing to send audio into KB-09's External Input.

Live 11 sidechain routing guide

Solve a problem

I cannot find KB-09 in my DAW

Extract the download, run INSTALL KB-09.cmd, and rescan VST3 plug-ins in your music software. Look under instruments. Use a 64-bit VST3 host. Logic requires a built AU, and ordinary Pro Tools requires a signed AAX release; use the separate unverified Mac download for AU. Signed AAX is unavailable.

I hear nothing

Load Calibrated Init, switch POWER on and play a fresh note. Check track monitoring, MIDI input, VOLUME and the DAW output. Close the EXT AUDIO overlay if it covers the keyboard. External presets need an audio source. In standalone, check audio/MIDI settings.

A sound keeps playing

VCA HOLD deliberately passes audio without a key. Try ENV or GATE, or load Calibrated Init. POWER mutes and clears note history; play a new note after turning it back on.

The window is large but blurry

Use the DAW-specific scaling setting . In Live, disabling Auto-Scale Plug-in Window fixed this for the owner. Then choose KB-09's own size. Do not lower the screen resolution to make the plug-in bigger.

The footer or a control is outside the window

Choose S or M, reopen the editor and check the host's scaling setting. Try a fresh instance after saving the patch. If it persists, note the DAW version, Windows display scale, monitor resolution and selected size when reporting the problem; a host can constrain a plug-in's requested dimensions.

Velocity, aftertouch, tempo or input trim does nothing

Select Modern. Authentic deliberately ignores these optional controls. Aftertouch also needs a controller that sends channel pressure. Onscreen mouse presses do not provide expressive aftertouch.

External audio is quiet or disappears

Confirm the host enabled the auxiliary input, raise EXT and use VCA Hold. Stereo sources with opposite-polarity channels cancel when folded to mono. In Modern, check INPUT TRIM. The EXT meter tracks activity; it is not a promise that your output path is audible.

Windows reports a missing VCRUNTIME or MSVCP file

Install Microsoft's current x64 Visual C++ Redistributable from Microsoft's official download page , then reopen the synth. Do not download individual DLLs from unrelated sites.

Microsoft's official download page

A missing cap, cracked key or broken bender looks alarming

These are visual AGE effects. The controls continue to work. Lower AGE to restore a fresher appearance; the same setting also returns the sound toward its healthy profile.

Take a test drive

Try these ideas as you get to know the instrument. Tick them as you go, and print the guide if you want to keep your notes. The browser's checkboxes reset when you reload.

Session notes

Date: _____ DAW / version: _____

Preset: _____ Favourite AGE: _____

Sounds, settings and ideas to return to:

Pitch-bender readout

Hover over or drag the spring-return bender to see its signed position, from -100% to +100%; centre is 0%. The VCO and VCF depth faders determine the resulting pitch and filter movement, so the position is not a fixed semitone amount. Release the bender to return to centre.

Help make KB-09 better

Please report bugs or mistakes in this guide to Kevin, the person who shared KB-09 with you. Include your operating system, music software and versions, what happened, steps to reproduce it, and a screenshot or exact error message. An AU download labeled unverified has not yet been confirmed in everyday Mac hosts; please report your Mac and host version when testing it.

Made for curiosity & playing

KB-09 is an unofficial personal recreation of Kevin's Roland SH-09, using a circuit-informed digital model and artwork modeled after his own aged instrument. “Authentic” names the original-style routing mode; individual hardware instruments can differ. This project is not affiliated with Roland Corporation and is not for sale.

The instrument has one voice, no onboard effects, no sequencer and no MIDI Learn. It reports 128 samples of processing latency. Hardware Gate/CV sockets, alternate tunings and per-note expressive protocols are not provided. Use your DAW for recording, effects and arrangement.

Open User Manual.pdf in the main download folder to read or print this guide. An HTML copy is retained in Support. Vendor documentation links need an internet connection.